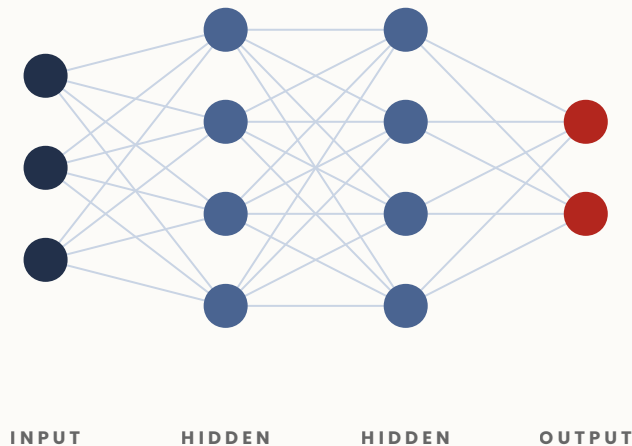


How a neural network learns, in one page

The loop I've taught to hundreds of MSc students — with zero equations.



Every line has a "volume knob" (a weight).
Learning = finding the right position for millions of knobs.

1

Guess

Data flows in, gets multiplied and mixed through the layers, an answer comes out. The first guesses are garbage — the knobs start random.

2

Measure the embarrassment

Compare the guess with the right answer. The difference is a single number — the **loss**. High loss = very wrong.

3

Assign blame backwards

Work backwards through the network asking: which knobs contributed most to the mistake? This is **backpropagation** — the whole trick of the field.

4

Nudge and repeat

Turn every knob a tiny bit in the blame-reducing direction. Repeat millions of times. That's it. That's the whole magic.

THE ONE-SENTENCE TAKEAWAY

A neural network is not programmed to be right — it is corrected into being less wrong, millions of times per hour.

From **Deep Learning Fundamentals** — the self-paced version of my university course. Full curriculum at brunomelicio.com/courses